

Profile

NTUST Chemical Engineering M.S. candidate holding three environmental specialist licences (**wastewater, waste, indoor air**), OSH exams scheduled; hands-on with **hazardous materials** (lithium metal, organic-solvent electrolytes, faulty Li-SOCl<sub>2</sub> cells), lab safety and **thermal hazard analysis** (TGA / DSC).

Experience

- 2025 – Present
- Sustainable Electrochemical Energy Development (SEED) Center, NTUST**  
Lab Equipment Procurement & Asset Maintenance Manager
  - Procurement (>NTD 2M):** Led procurement of glovebox O<sub>2</sub> sensors, a DSC press set with annual maintenance contract, a hydraulic electrode punch, vacuum pumps, 10 Eppendorf pipettes and low-temperature baths.
  - Lab safety:** monitor glovebox O<sub>2</sub> / moisture levels and handle sensor faults; manage and inspect gas cylinders, run lab safety walk-throughs and brief newcomers on safety; helped with on-site response to incidents such as a cut from broken glassware and a glovebox catalyst-canister leak.
  - Hazardous materials:** handle lithium metal and assemble coin cells in a glovebox, prepare ether / carbonate organic-solvent electrolytes, and dismantle and handle faulty Li-SOCl<sub>2</sub> cells.
  - TGA testing service:** Ran TGA measurements and data interpretation for graduate students from 6 labs across Chemical, Mechanical and Materials Engineering at NTUST.
  - Instrument maintenance:** Responsible for maintenance and calibration of the key thermal analysis instruments, TGA Q500 and DSC 2500.
- Summer 2022
- New Taipei City Dept. of Urban Development**  
Summer Intern, Housing Development Section
  - Reviewed rental subsidy applications and indexed confidential records in government databases; handled frontline public inquiries, building strong familiarity with administrative workflows and SOPs.
- 2021 – 2024
- Tunghai University Student Housing / GEPT**  
Resident Assistant (2 yrs) / Exam Staff (part-time)
  - Oversaw safety and daily order in a 100-resident dormitory, mediating resident disputes and liaising with university administration; as part-time exam staff, handled test-paper check-in with zero errors.

Research & Industry

**Long-Life Li Battery Evaluation for Smart Water Meters**  
Partner: a smart-water-meter battery maker **Project lead**

**Screening methodology**  
Designed full-stage test protocols, experimental workflows and statistical validation models.

**Database & SOP**  
Established the test database and standardized high-throughput screening procedures (SOP).

**Advanced analysis**  
EIS, OCV drip, self-discharge / lifetime estimation, accelerated aging and X-ray CT internal structure analysis.

**Training & tech transfer**  
Trained QA and R&D teams to independently run large-batch screening and data interpretation.

**Nature Energy** Journal submission — battery thermal stability study Status **Under review**  
**Co-author** Responsible for TGA thermal measurement  
Performed TGA sample preparation, measurement setup and data interpretation for battery material samples, providing thermal stability data supporting the manuscript's conclusions.

Conferences & Training

- Presenter** 2026 TwICHe Annual Meeting  
2026/12/05-06 • Feng Chia University  
Title: *Weakly Solvating Allyl Ether Electrolytes for Subzero Li || SPAN Batteries: Solvation Structure, Transport Properties, and Separator Wettability*
- Vendor seminar** TA Instruments ARES-G3 launch & applications forum  
2026/07/17 • National Taiwan University
- Vendor lecture** TA Instruments TGA lecture  
2026/06/03 • National Taiwan University

EHS Licenses

- Dedicated Wastewater and Sewage Treatment Specialist (Class B)**  
Certified 2026
- Indoor Air Quality Maintenance Management Specialist**  
Certified 2026
- Waste Disposal Technician (Class A)**  
Certified 2026
- Occupational Safety and Health Management Personnel (Level B)**  
2026 / 11 • mid-Nov exam scheduled
- Occupational Safety Management Specialist (Level A)**  
2027 / 03 • exam scheduled

Education

- 2025 – 2027 (expected)  
**National Taiwan University of Science and Technology**  
**M.S. in Chemical Engineering**  
Focus on low-temperature electrolytes and battery aging. Coursework: Nanomaterials and Applications (lithography, nanoimprint & self-assembly), Process Safety & Radiation Protection.
- 2020 – 2024  
**Tunghai University**  
**B.S. in Chemical & Materials Engineering**  
Focus on unit operations & process control (Aspen Plus V14 / Aspen Dynamics simulation).

Skills & Instruments

**EHS fundamentals**

Chemical labelling rules (GHS)

Explosive limits (LEL / UEL)

Occupational exposure limits

Ventilation

**Electrochemistry & batteries**

Impedance spectroscopy (EIS)

Battery aging analysis

Reliability analysis

Low-temp electrolytes

**Process & operations**

Lithography (nanoimprint / self-assembly)

Equipment procurement & maintenance

**Software**

Aspen Plus V14 Aspen Dynamics

Microsoft Office OriginPro

TA Universal Analysis TA TRIOS

Instruments

- TGA Q500 Maintenance
- DSC 2500 Maintenance
- ARES-G3 Operation
- Glovebox O<sub>2</sub> sensor Procurement
- DSC sample press Procurement + service
- Hydraulic electrode punch Procurement
- Vacuum pump Procurement
- Eppendorf Pipette Procurement
- Low-temperature bath Procurement